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[4657]-578

S.E. (Computer) (Second Semester) EXAMINATION, 2014

COMPUTER GRAPHICS AND GAMING

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Attempt Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6 and Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn whenever necessary.

(iii) Assume suitable data, if necessary.

1. (a) What is computer graphics ? State the applications of computer graphics. [6]
- (b) Explain Bresenham's line algorithm and find out which pixels would be turned on for the line with end points (5, 2) to (8, 4) using the same. [6]

Or

2. (a) Explain TIFF file organization with block diagram. [6]
- (b) What is aliasing and anti-aliasing ? List and explain 2 anti-aliasing techniques. [6]

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3. (a) Explain the different methods for testing a pixel inside a polygon. [5]
- (b) Explain the concept of 3D rotation about an arbitrary axis with an example. [7]

Or

4. (a) Write transformation matrix for Scaling and Rotation and scale the polygon with co-ordinates A(4, 5), B(8, 10) and C(8, 2) by 2 units in x-direction and 3 units in y-direction. Find the transformed A, B and C points. [6]
- (b) Explain Sutherland-Hodgeman clipping algorithm with example. [6]
5. (a) Explain the light, reflectivity, color and shading in computer graphics. [5]
- (b) Define Bezier curve. State its properties. Derive blending function of Bezier curve. [8]

Or

6. (a) Explain and compare shading algorithms. [6]
- (b) Describe any *two* hidden face removal algorithm with diagram. [7]

7. (a) What is a segment ? Give its structure and also describe various operations carried out on the segment. [7]

(b) Write a short note on 3D MaxStudio or Maya. [6]

Or

8. (a) Draw block diagram of i860. [6]

(b) What is the difference between conventional and computer based animations ? What are the various methods of controlling animation ? [7]